

User manual for an electronic load



A master thesis by

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1. Introduction

Purpose of device:

The advanced electronic load system has been designed to simulate real-world operating conditions for fuel cell stack characterization. Its main purpose is to enable precise control and measurement of Fuel cell, monitor stack parameters such as cell voltages, current, power, temperature, and to log all data for analysis.

This system supports operating mode such as:

- Constant Current (CC)
- Constant Voltage (CV)

Additionally, the integrated data logging, real-time display, valve control, blower and fan management, and persistent memory settings make it ideal for laboratory environments focusing on fuel cell research and performance evaluation.

System specifications:

<i>Parameter</i>	<i>Specification</i>
Nominal Input Voltage	0 ... 13 V
Max Input Voltage	15 V
Max Input Current	150 A
Operating Modes	CC / CV
Cell Monitoring	Up to 12 individual cells
Nominal Voltage Range per cell	0 ... 1.1 V
Max Input voltage per cell	4.5 V
Voltage Tolerance	± 3 mV
Data Logging Interval	Fixed at 0.5 seconds
Supply Voltage of Device	24 V
Max Supply Current	7 A
Supply voltage (Valves, Blower, Pump)	12 V / 24 V (select by solder jumpers)
Microcontroller	ESP32-PICO-D4
SD Card Support	Yes (formatted FAT32)
Software Version	2.3.4
Firmware Update	Via USB using Arduino IDE (v2.3.4, ESP32 2.0.17)

2. Safety Instructions

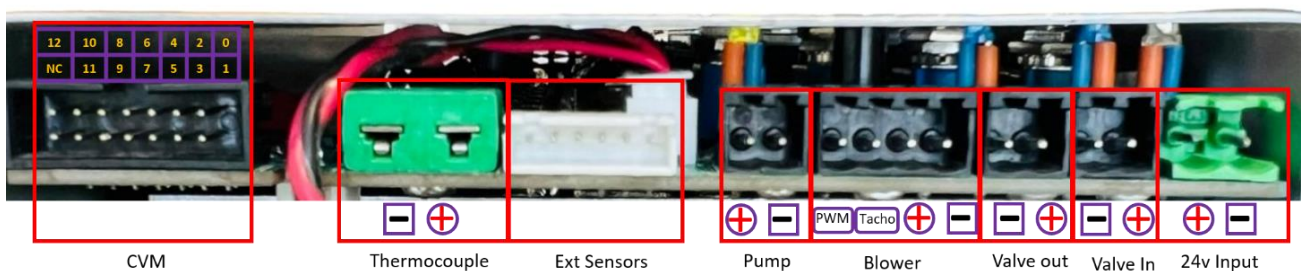
- Ensure correct polarity and voltage levels when connecting power sources or test loads.
- Never exceed the rated current or voltage specifications.
- Ensure the cooling fan and blower vents are unobstructed during use.
- Do not insert or remove the SD card while the device is **ON**.
- Ensure all external connections are made with power **OFF** to prevent damage to components.
- Do not connect the fuel cell stack to the load when power is **ON**.

3. Hardware Overview

The below figure represents the hardware overview.



Detail overview of the Fuel cell peripherals Connectors

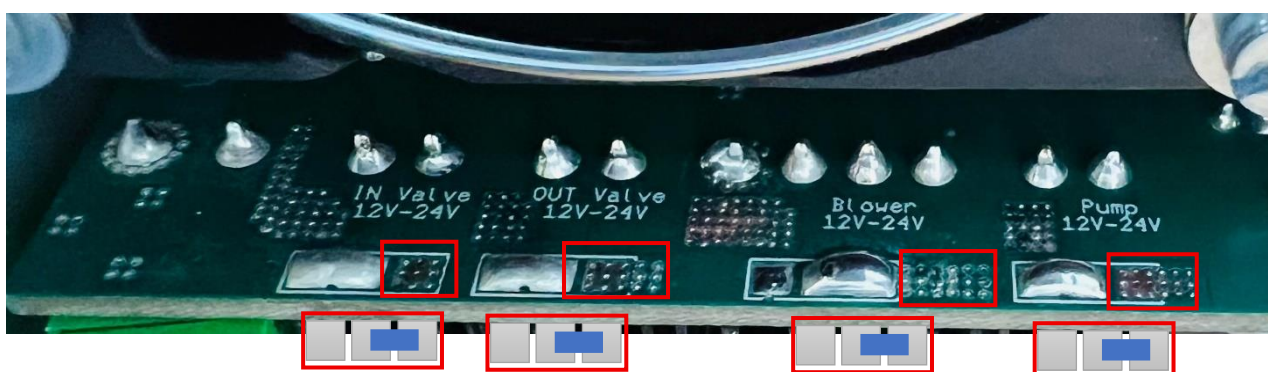


<i>Label</i>	<i>Description</i>
FC In (+ / -)	Main input terminal for connecting the Fuel Cell. Ensure correct polarity.
GND	Common ground terminal shared with FC input.
24V In	Power input connector for supplying the device (typically 24V DC).
Valve In / Valve Out	Control pins for operating purge and input valves.
Blower	Connector for blower fan (air management).
Pump	Connector for operating coolant pump.
Ext Sensors	Input for connecting additional external temperature or voltage sensors.
Temp FC	Analog input for fuel cell temperature sensor (Thermocouple).
CVM (Cell Voltage Monitor)	12-channel cell voltage measurement connector.

Valve, Blower, and Pump Voltage Configuration Interface:

The PCB shown provides output control terminals for: IN Valve, Purge Valve, Blower and Pump with (12V–24V)

Currently, the components are configured to operate with a 12V supply. To switch to a 24V supply, the soldered jumpers must be desoldered from the 12V terminal and resoldered to the adjacent 24V terminal as indicated on the bottom layer of PCB silkscreen as shown in below figure.



4. Installation and Setup

1. Connect the PC and ESP32 using USB to upload the latest code.
2. Connect fuel cell positive and Negative to the **In** and **GND** bus of the fuel cell.
3. Connect Valve In, Valve Out, Pump, and Blower to their respective terminals.
4. Connect CVM Module to the fuel cell stack.
5. Insert a FAT32-formatted microSD card into the SD card slot for data logging.
6. Provide a stable 24V DC power input to the board through the 24V In terminal.
7. Use the rotary encoder to navigate the menu, set operating parameters, and start the modes.

5. Operating Instructions

1. Powering On

- Ensure all connections are properly made (as per Installation)
- Turn on the 24V DC supply to power the device.
- Turn on the switches of valves, blower and load to **Auto mode**.
- Wait for the display to initialize.

2. Navigating the Menu

- Use the **rotary encoder** to scroll the yellow highlighter through menu items.
- Press the encoder to **select** a setting or action.

Key display sections:

- **Main Screen:** Shows FC voltage, FC current, power, charge, energy, temperature, pump status, and Air %.
- **Bottom Menu:** Options for Mode selection, Start/Stop, and entering the settings menu.

The below figure represents the detailed Info about the GUI.



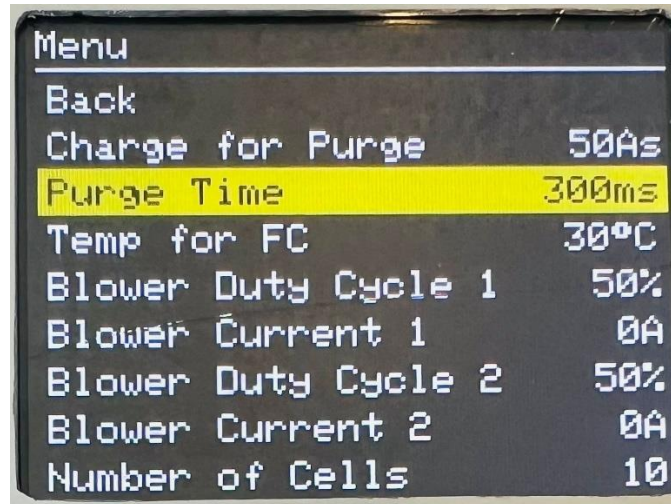
3. Setting Parameters

Go to Menu and select parameters like:

Parameter	Minimum Value	Maximum Value	Unit
Charge for Purge	1	200	AS (Ampere-seconds)
Purge Time	100	500	ms (milliseconds)
Temperature for FC	30	120	°C (Celsius)
Blower Duty Cycle 1	50	100	% (Percentage)
Blower Current 1	0.0	10.0	A (Amperes)
Blower Duty Cycle 2	50	100	% (Percentage)
Blower Current 2	0.0	10.0	A (Amperes)
Number of Cells	1	12	-

Operating Instructions - Accessing and Adjusting Parameters

1. The highlighter (yellow Rectangle) is initially set to **START** when the device is powered on.
2. To access the parameters in Menu:
 - Rotate the encoder three times **clockwise**.
 - This will bring you to the Menu.
3. To select a parameter:
 - Press the encoder button to enter the menu.
 - Rotate the encoder to navigate through the list of parameters.
 - When you reach the parameter, you want to adjust, press the encoder button.
4. When the parameter is selected:
 - You will see a yellow fill around the parameter, indicating it is ready for adjustment as seen in below figure.



5. To adjust the value:

- Rotate the encoder to increase or decrease the value as needed.

6. To save the adjusted value:

- Press the encoder button again.
- This will save the value and bring you back to the main menu.

7. You are now ready for the next selection or operation.

4. Selecting Operating Mode:

Choose the desired mode using the **Mode** option:

CC – Constant Current mode

CV – Constant Voltage mode

Operating Instructions - Selecting Mode and Setting I-set

To select CC mode:

1. When you power on the electronic load, it automatically starts in the previously selected mode.

- For example, if it was in Constant Current (CC) mode last time, it will remain in CC mode upon startup.

2. To set the I-set for CC mode:

- Rotate the encoder **counterclockwise** until you reach I-set.
- Press the encoder button to select it.
- Rotate the encoder to adjust the current to your desired value.

- Press the encoder button again to **save** the value.



3. To return to Start:

- Rotate the encoder clockwise until the highlighter reaches Start and select it so the device will operate in CC mode.

To select CV mode:

If you want to switch to Constant Voltage (CV) mode, simply rotate the encoder **clockwise**. The highlighter will move to CV. Press the encoder button to confirm the selection, and the device will switch to CV mode.

5. Starting and stopping operation

- Move the cursor to **Start** and press the encoder to begin.
- Live data will be updated on the display.
- Data is logged to the SD card every 0.5 seconds.
- To stop operation, select **Stop** and press the encoder.

6. Viewing logged data

- Remove the SD card after stopping the test. (only when the load is in **OFF**)
- Insert into a PC to access .txt files with headers and time-series data.